Maths for EE

CBSE 2015 Combined

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers. The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. Class 10 2015

A. Formulae

1.1.1

$$p_{k+1} = p_k - \frac{F(p_k)}{F'(p_k)} \tag{1.1.1.1}$$

1.1.2

$$P(X = i, Y = j) = P(X = i)P(Y = j) \tag{1.1.2.1}$$

$$P(XY = z) = \sum_{ij=z} P(X = i, Y = j). \tag{1.1.2.2}$$

1.1.3

$$S_n = \frac{n}{2}\big(2a + (n-1)d\big). \tag{1.1.3.1}$$

$$Au = b. \tag{1.1.3.2}$$

$$u^{(k+1)} = D^{-1}\big[b - (L+U)u^{(k)}\big]. \tag{1.1.3.3}$$

$$B = -D^{-1}(L+U) \tag{1.1.3.4}$$

1.1.5

$$X(z) = \sum_{n=1}^{\infty} x[n]z^{-n} = \frac{az^{-1}}{1-z^{-1}} + \frac{dz^{-1}}{(1-z^{-1})^2}. \tag{1.1.4.1}$$

1.1.6

$$P(R) + P(B) + P(O) = 1. \tag{1.1.5.1}$$

1.1.8

$$\mathbf{1}_{S\cup A} = \mathbf{1}_S + \mathbf{1}_A - \mathbf{1}_{S\cap A}. \tag{1.1.6.1}$$

1.1.9

$$2[ABC] = |\det(B-A, C-A)| \tag{1.1.7.1}$$

B. Problems

1.2.1 If the quadratic equation $px^2 - 2\sqrt{5}\,px + 15 = 0$ has two equal roots, find p.

Solution:

For a repeated root, the two roots coincide precisely when the discriminant vanishes. Thus

$$\Delta = (-2\sqrt{5}p)^2 - 4(p)(15) = 20p^2 - 60p = 20p(p-3) = 0.$$

The algebraic candidates are $p = 0$ and $p = 3$. The first candidate does not define a quadratic equation: its left side becomes 15, which cannot be zero. Therefore the admissible parameter is

$$p = 3.$$

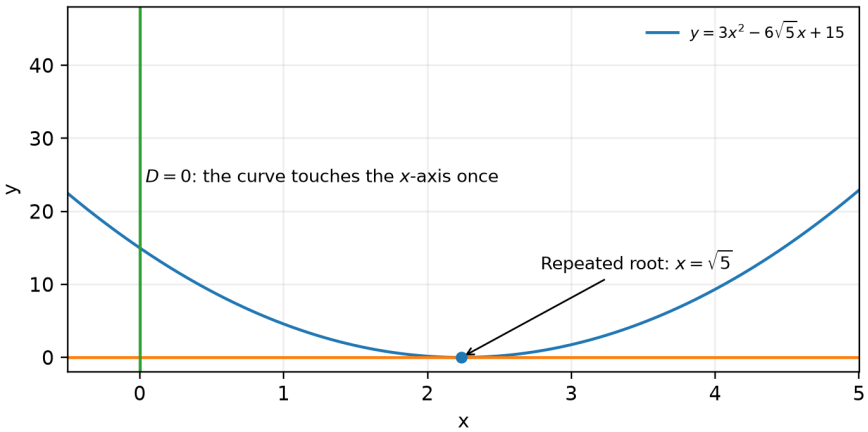
As an engineering root-finding check, set $F(p) = p(p-3)$. Newton–Raphson gives

$$p_{k+1} = p_k - \frac{F(p_k)}{F'(p_k)} \tag{1.1.1.1}$$

$$= p_k - \frac{p_k(p_k-3)}{2p_k-3}.$$

At the desired simple root $p = 3$, $F'(3) = 3 \neq 0$ and the Newton map has the expected local quadratic convergence; the singular midpoint $p = 3/2$ must be avoided.

$$p = 3$$



Code reference -

codes/C10/Q01.py

1.2.2 Two different dice are tossed together. Find the probability that the product of their top numbers is 6.

Solution:

Let X and Y be the independent outcomes of the two fair dice. Their joint mass function is

$$P(X=i,Y=j) = P(X=i)P(Y=j) \tag{1.1.2.1}$$
$$= \frac{1}{36}, \qquad 1 \le i,j \le 6.$$

The product is a discrete push-forward random variable. Its probability is therefore obtained by summing the joint mass over the product preimage:

$$P(XY=z) = \sum_{ij=z} P(X=i,Y=j). \tag{1.1.2.2}$$

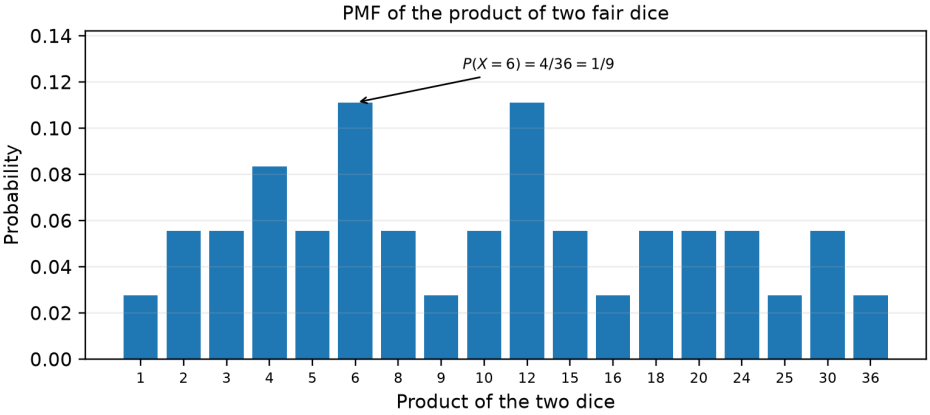
For $z=6$, the admissible ordered pairs are

$$\{(i,j) : ij = 6\} = \{(1,6),(2,3),(3,2),(6,1)\}.$$

There are four equally likely pairs out of $6 \times 6 = 36$, so

$$P(XY=6) = 4\left(\frac{1}{36}\right) = \frac{4}{36} = \frac{1}{9}.$$

The four pairs exhaust the positive factorisations of 6 within the die support, which verifies that no outcome has been omitted.



Code reference -

`codes/C10/Q03.py`

1.2.3 In an arithmetic progression, $S_5 + S_7 = 167$ and $S_{10} = 235$. Find the AP.

Solution:

For an AP with first term a and common difference d,

$$S_n = \frac{n}{2}\big(2a + (n-1)d\big). \tag{1.1.3.1}$$

Consequently,

$$S_5 + S_7 = (5a + 10d) + (7a + 21d) = 12a + 31d.$$

$$S_{10} = 10a + 45d.$$

Writing $u = (a, d)^T$, the engineering linear-system model is

$$Au = b. \tag{1.1.3.2}$$

$$A = \begin{pmatrix} 12 & 31 \\ 10 & 45 \end{pmatrix}.$$

$$b = \begin{pmatrix} 167 \\ 235 \end{pmatrix}.$$

Use the Jacobi splitting $A = D + L + U$, where

$$D = \begin{pmatrix} 12 & 0 \\ 0 & 45 \end{pmatrix}.$$

$$L + U = \begin{pmatrix} 0 & 31 \\ 10 & 0 \end{pmatrix}.$$

The iteration and its matrix are

$$u^{(k+1)} = D^{-1}\big[b - (L + U)u^{(k)}\big]. \tag{1.1.3.3}$$

$$B = -D^{-1}(L + U) \tag{1.1.3.4}$$

$$= \begin{pmatrix} 0 & -31/12 \\ -2/9 & 0 \end{pmatrix}.$$

Since $\rho(B) = \sqrt{31/54} < 1$, the iteration converges to the unique fixed point. Solving the fixed-point equations gives

$$12a + 31d = 167.$$

$$10a + 45d = 235.$$

and elimination yields $115d = 575$, so $d = 5$ and then $a = 1$. Hence

$$1, 6, 11, 16, \ldots.$$

Finally, $S_5 = 5{+}10(5) = 55$ and $S_7 = 7{+}21(5) = 112$, while $S_{10} = 10{+}45(5) = 235$. The residual vector is therefore $(0, 0)^T$.

Code reference -

codes/C10/Q08.py

1.2.4 If $A(x, y)$, $B(-5, 7)$ and $C(-4, 5)$ are collinear, find the relation between x and y.

Solution:

Collinearity means that the two displacement vectors from A are linearly dependent.

Their determinant must therefore vanish:

$$\det\begin{pmatrix} -5-x & 7-y \\ -4-x & 5-y \end{pmatrix} = 0.$$

Subtracting the first row from the second gives the equivalent translated determinant

$$\det\begin{pmatrix} x+5 & y-7 \\ 1 & -2 \end{pmatrix} = 0.$$

Expanding along the first row,

$$-2(x+5) - (y-7) = 0 \Longrightarrow -2x - y - 3 = 0.$$

Thus the required rank-deficiency relation is

$$2x + y + 3 = 0.$$

1.2.5 The 14th term of an AP is twice its 8th term. If the 6th term is -8, find the sum of its first 20 terms.

Solution:

Model the AP as the discrete-time sequence $x[n] = a + (n-1)d$, $n \geq 1$. Its unilateral transform separates the constant and ramp components:

$$X(z) = \sum_{n=1}^{\infty} x[n] z^{-n} = \frac{az^{-1}}{1-z^{-1}} + \frac{dz^{-1}}{(1-z^{-1})^2}. \tag{1.1.4.1}$$

The term conditions give two linear equations:

$$a + 13d = 2(a + 7d), \qquad a + 5d = -8.$$

The first equation reduces to $a + d = 0$, so $a = -d$. Substitution into the second equation gives

$$a + 13d = 2(a + 7d).$$

$$a + 5d = -8.$$

The twentieth partial sum is consequently

$$S_{20} = \frac{20}{2}\big(2a + 19d\big) = 10(4 - 38) = -340.$$

The sequence generated by $(a, d) = (2, -2)$ has $t_6 = -8$ and $t_{14} = -24 = 2t_8$, confirming both data constraints.

Code reference -

codes/C10/Q11.py

1.2.6 A jar contains only red, blue and orange balls. The probabilities of drawing a red and a blue ball are $\frac{1}{4}$ and $\frac{1}{3}$, respectively. If there are 10 orange balls, find the total number of balls.

Solution:

Let N be the total number of balls. The red, blue, and orange counts must be nonnegative integers, and probability conservation gives

$$P(R) + P(B) + P(O) = 1. \tag{1.1.5.1}$$

$$\frac{1}{4} + \frac{1}{3} + \frac{10}{N} = 1.$$

Combining the first two fractions,

$$\frac{1}{4} + \frac{1}{3} = \frac{7}{12}.$$

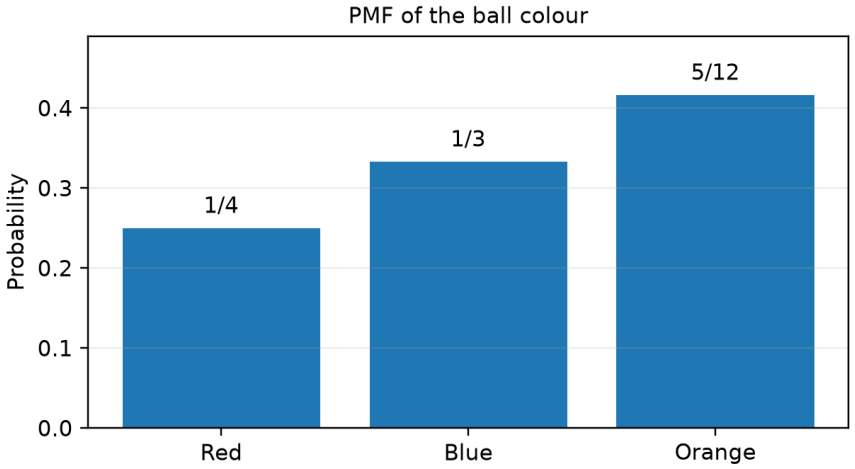
$$\frac{10}{N} = 1 - \frac{7}{12} = \frac{5}{12}.$$

Therefore $N = 10(12/5) = 24$. The implied counts are

$$R = \frac{N}{4} = 6, \qquad B = \frac{N}{3} = 8, \qquad O = 10,$$

and $R + B + O = 6 + 8 + 10 = 24$. Thus the integer feasibility check agrees with

$$N = 24.$$



Code reference -

codes/C10/Q15.py

1.2.7 Find the 60th term of the AP $8, 10, 12, \ldots$ and the sum of its last 10 terms.

Solution:

Index the finite AP from zero so that $x[n] = 8 + 2n$ for $0 \le n \le 59$. The finite-support

signal, rather than an infinite extension, is represented by

$$X(z) = 8\sum_{n=0}^{59} z^{-n} + 2\sum_{n=0}^{59} nz^{-n}.$$

The requested term is the sample at the final index:

$$x[59] = 8 + 2(59) = 126.$$

The last ten terms are indexed by $n = 50, \ldots, 59$, with

$$x[50] = 8 + 100 = 108.$$

$$x[59] = 126.$$

Using the arithmetic-series sum,

$$\sum_{n=50}^{59} x[n] = \frac{10}{2}(108 + 126) = 5(234) = 1170.$$

The finite index range confirms that no samples beyond the 60-term support have entered the calculation.

Code reference -

codes/C10/Q22.py

1.2.8 A card is drawn from a well-shuffled 52-card deck. Find the probability that it is (i) a spade or an ace, (ii) a black king, (iii) neither a jack nor a king, (iv) either a king or a queen.

Solution:

Use indicator variables to preserve the overlap correction in unions. For a spade S and an ace A, $|S| = 13$, $|A| = 4$, and $|S \cap A| = 1$. Thus

$$\mathbf{1}_{S\cup A} = \mathbf{1}_S + \mathbf{1}_A - \mathbf{1}_{S\cap A}. \tag{1.1.6.1}$$

$$P(S \cup A) = \frac{|S| + |A| - |S \cap A|}{52}.$$

The four requested probabilities follow from the corresponding event counts:

$$P(S \cup A) = \frac{13 + 4 - 1}{52} = \frac{4}{13}.$$

$$P(\text{black king}) = \frac{2}{52} = \frac{1}{26}.$$

$$P(\text{neither jack nor king}) = 1 - \frac{4 + 4}{52} = \frac{44}{52} = \frac{11}{13},$$

and, since kings and queens are disjoint,

$$P(K \cup Q) = \frac{4 + 4}{52} = \frac{2}{13}.$$

Hence the answers are

$$\frac{4}{13},\quad \frac{1}{26},\quad \frac{11}{13},\quad \frac{2}{13}.$$

1.2.9 Find k if the triangle with vertices $(1,-1)$, $(-4,2k)$ and $(-k,-5)$ has area 24 square units.

Solution:

The determinant of the two edge vectors gives twice the signed area. With $A=(1,-1)$, $B=(-4,2k)$, and $C=(-k,-5)$,

$$2[ABC] = |\det(B-A, C-A)| \tag{1.1.7.1}$$

$$= \left|\det\begin{pmatrix} -5 & 2k+1 \\ -k-1 & -4 \end{pmatrix}\right| = |2k^2+3k+21|.$$

The inner quadratic is always positive because its discriminant is $3^2-4(2)(21) = -159 < 0$ and its leading coefficient is positive. Therefore the absolute value can be removed, and the area condition becomes

$$2k^2+3k+21 = 48 \Longrightarrow 2k^2+3k-27=0.$$

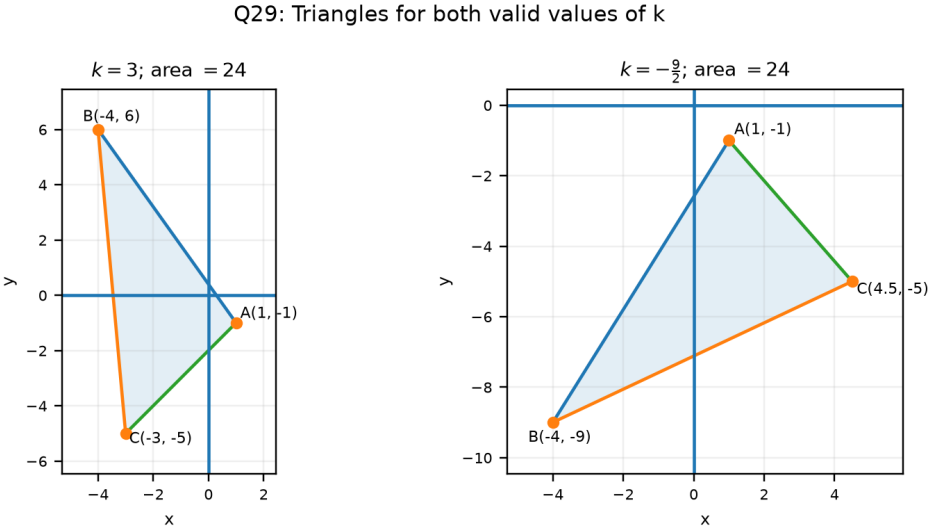
Applying the quadratic formula,

$$k = \frac{-3\pm\sqrt{9+216}}{4} = \frac{-3\pm 15}{4}.$$

Thus both feasible parameter values are

$$k=3 \quad \text{or} \quad k=-\frac{9}{2}.$$

Each value gives determinant magnitude 48, hence area 24.



Code reference -

codes/C10/Q29.py

2. Class 12 2015

2.1.1 $$\mathrm{Null}(M) = \{\mathbf{v} : M\mathbf{v} = 0\}. \tag{2.1.1.1}$$

2.1.2 $$\mathbf{d} = t(\mathbf{a} \times \mathbf{b}) \tag{2.1.2.1}$$

2.1.3 $$P_\perp = I - \frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}. \tag{2.1.3.1}$$

$$d = \|P_\perp \mathbf{w}\| = \frac{\|\mathbf{w} \times \mathbf{v}\|}{\|\mathbf{v}\|}. \tag{2.1.3.2}$$

2.1.4 $$(\mathbf{n}_1 + \lambda\mathbf{n}_2) \cdot \mathbf{d} = 0. \tag{2.1.4.1}$$

2.1.5 $$\mathbf{r} = \mathbf{A} + t\mathbf{d}. \tag{2.1.5.1}$$

2.1.6 $$\frac{d}{dx} \log |u| = \frac{u'}{u}. \tag{2.1.6.1}$$

$$\theta y - y = xy' - y. \tag{2.1.6.2}$$

2.1.7 $$\det(A^n) = (\det A)^n. \tag{2.1.7.1}$$

2.1.8 $$AA^{-1} = I. \tag{2.1.8.1}$$

2.1.9 $$\lambda_0 = a + b + c. \tag{2.1.9.1}$$

$$\lambda_1 = a + b\omega + c\omega^2. \tag{2.1.9.2}$$

$$\lambda_2 = a + b\omega^2 + c\omega. \tag{2.1.9.3}$$

$$0 = \det A = -\lambda_0\lambda_1\lambda_2 = -(a+b+c)(a^2+b^2+c^2-ab-bc-ca). \tag{2.1.9.4}$$

$$a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}\big((a-b)^2+(b-c)^2+(c-a)^2\big). \tag{2.1.9.5}$$

2.1.12

$$(a,b)*(c,d) = (a+c, b+d). \tag{2.1.10.1}$$

$$\begin{aligned}((a,b)*(c,d))*(e,f) &= (a+c+e, b+d+f)\\ &= (a,b)*((c,d)*(e,f)).\end{aligned} \tag{2.1.10.2}$$

$$(a,b)*(c,d) = (a+c,b+d) = (c+a,d+b) = (c,d)*(a,b). \tag{2.1.10.3}$$

2.1.13

$$a*b = (a+b) \bmod 7. \tag{2.1.11.1}$$

$$a*(7-a) = a+7-a \equiv 0 \pmod 7. \tag{2.1.11.2}$$

2.1.14

$$x - x^2 = x(1-x). \tag{2.1.12.1}$$

2.1.16

$$P(H_2 \mid E) = \frac{P(E \mid H_2)P(H_2)}{P(E)}. \tag{2.1.13.1}$$

2.1.17

$$G_X(z) = \sum_x P(X=x)z^x. \tag{2.1.14.1}$$

2.1.18

$$G(z) = (1-p+pz)^3. \tag{2.1.15.1}$$

2.1.19

$$P(E) = \sum_i P(E \mid S_i)P(S_i). \tag{2.1.16.1}$$

B. Problems

2.2.1 Find a vector of magnitude $\sqrt{171}$ perpendicular to both $\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$.

Solution:

Perpendicularity requires a vector in the null space of the matrix with rows $\mathbf{a}^T$ and $\mathbf{b}^T$:

$$M = \begin{pmatrix} 1 & -2 & -3 \\ 3 & -1 & -2 \end{pmatrix}.$$

$$M\mathbf{d} = 0.$$

The two rows are independent, so rank–nullity gives a one-dimensional null space. Its antisymmetric cross-product basis is

$$\text{Null}(M) = \{\mathbf{v} : M\mathbf{v} = 0\}. \tag{2.1.1.1}$$

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 1 \\ -7 \\ 5 \end{pmatrix}.$$

Indeed,

$$\mathbf{a} \cdot (1, -7, 5) = 1 + 14 - 15 = 0.$$

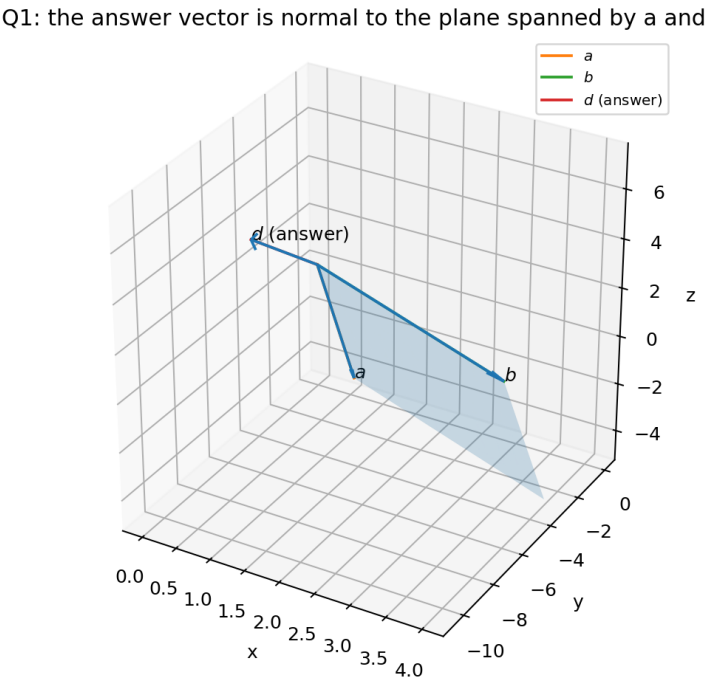
$$\mathbf{b} \cdot (1, -7, 5) = 3 + 7 - 10 = 0.$$

Every admissible vector is therefore $\mathbf{d} = t(1, -7, 5)^T$. Since

$$\|(1, -7, 5)\|^2 = 1 + 49 + 25 = 75,$$

the magnitude condition gives $75t^2 = 171$, or $t = \pm\sqrt{57}/5$. Hence

$$\mathbf{d} = \pm\frac{\sqrt{57}}{5}\begin{pmatrix} 1 \\ -7 \\ 5 \end{pmatrix}.$$



Code reference -

codes/C12/Q01.py

2.2.2 Let $\mathbf{a} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 3 \\ -2 \\ 7 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$. Find $\mathbf{d}$ perpendicular to both $\mathbf{a}, \mathbf{b}$ with $\mathbf{c} \cdot \mathbf{d} = 27$.

Solution:

The antisymmetric bilinear operation supplies a direction perpendicular to both prescribed vectors. Computing its determinant gives

$$\mathbf{d} = t(\mathbf{a} \times \mathbf{b}) \tag{2.1.2.1}$$

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 32 \\ -1 \\ -14 \end{pmatrix}.$$

Thus the complete perpendicular family is

$$\mathbf{d} = t \begin{pmatrix} 32 \\ -1 \\ -14 \end{pmatrix}.$$

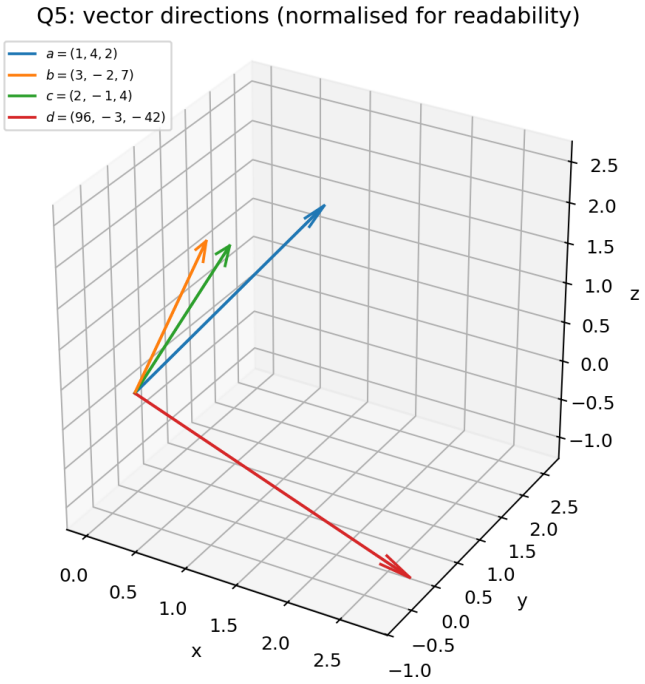
The remaining scalar constraint selects the scale. First,

$$\mathbf{c} \cdot (32, -1, -14) = 2(32) + (-1)(-1) + 4(-14) = 64 + 1 - 56 = 9.$$

Therefore $9t = 27$, so $t = 3$ and

$$\mathbf{d} = \begin{pmatrix} 96 \\ -3 \\ -42 \end{pmatrix}.$$

The dot products $\mathbf{a} \cdot \mathbf{d} = 0$ and $\mathbf{b} \cdot \mathbf{d} = 0$, together with $\mathbf{c} \cdot \mathbf{d} = 27$, verify the corrected result.



Code reference -

`codes/C12/Q05.py`

2.2.3 Find the shortest distance between $\mathbf{r} = (1, 2, 3) + \lambda(2, 3, 4)$ and $\mathbf{r} = (2, 4, 5) + \mu(4, 6, 8)$.

Solution:

The direction vectors $(2, 3, 4)$ and $(4, 6, 8)$ are parallel. Choose one point on each line and form the joining vector

$$\mathbf{v} = (2, 3, 4)^T, \qquad \mathbf{w} = (2, 4, 5) - (1, 2, 3) = (1, 2, 2)^T.$$

Only the component of $\mathbf{w}$ orthogonal to the common direction contributes to the shortest

distance. The projection-residual operator is

$$P_{\perp} = I - \frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}. \tag{2.1.3.1}$$

$$d = \|P_{\perp}\mathbf{w}\| = \frac{\|\mathbf{w} \times \mathbf{v}\|}{\|\mathbf{v}\|} \tag{2.1.3.2}$$

Here

$$\mathbf{w} \times \mathbf{v} = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}.$$

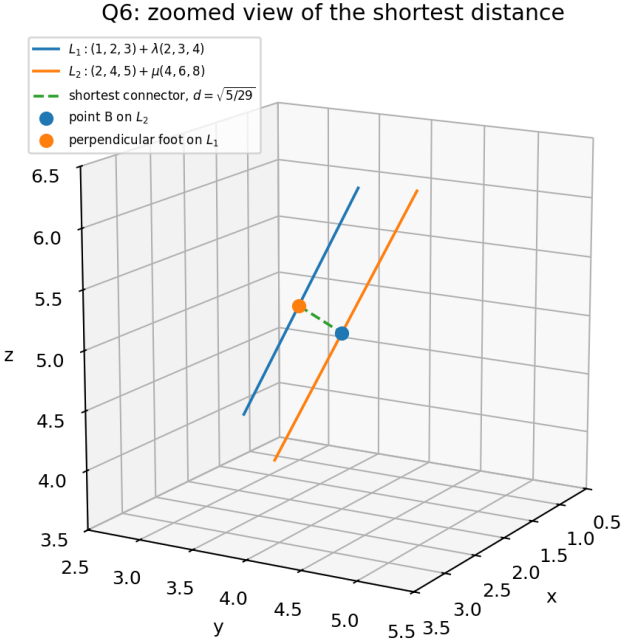
$$\|\mathbf{w} \times \mathbf{v}\| = \sqrt{5}.$$

$$\|\mathbf{v}\| = \sqrt{29}.$$

Consequently,

$$d = \sqrt{\frac{5}{29}}.$$

The cross product is orthogonal to $\mathbf{v}$, so the residual satisfies the required projection consistency check.



Code reference -

codes/C12/Q06.py

2.2.4 Find the plane through the intersection of $2x + y - z = 3$ and $5x - 3y + 4z + 9 = 0$ that is parallel to $\frac{x-1}{2} = \frac{y-3}{-4} = \frac{5-z}{-5}$.

Solution:

Let

$$P_1 = 2x + y - z - 3.$$

$$P_2 = 5x - 3y + 4z + 9.$$

Every plane through the intersection line belongs to the affine pencil $P_1 + \lambda P_2 = 0$. Its normal is

$$\mathbf{n}(\lambda) = (2 + 5\lambda,\ 1 - 3\lambda,\ -1 + 4\lambda).$$

The direction of the given line is $\mathbf{d} = (2, -4, 5)$. The required parallelism condition is

$$(\mathbf{n}_1 + \lambda \mathbf{n}_2) \cdot \mathbf{d} = 0. \tag{2.1.4.1}$$

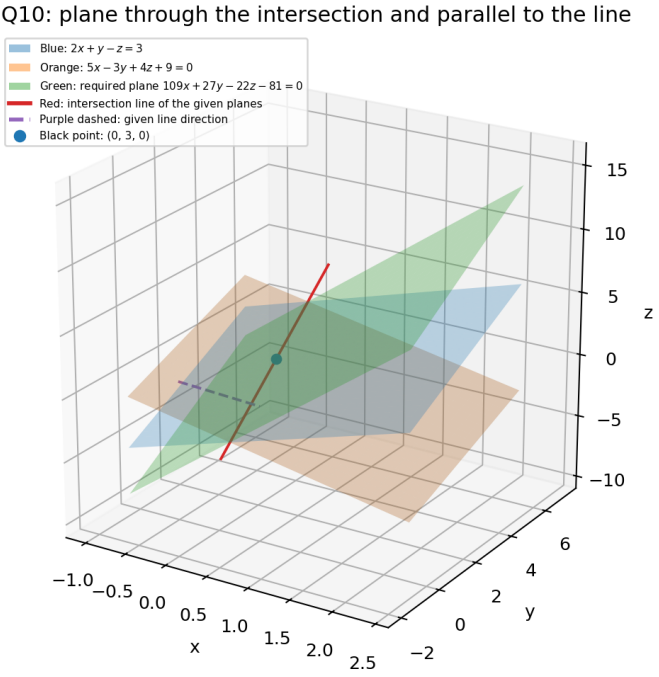
Substitution gives

$$2(2 + 5\lambda) - 4(1 - 3\lambda) + 5(-1 + 4\lambda) = 0,$$

so $-5 + 42\lambda = 0$ and $\lambda = 5/42$. Multiplying the pencil by 42 yields

$$42P_1 + 5P_2 = 0 \Longrightarrow 109x + 27y - 22z - 81 = 0.$$

The plane contains the intersection line by construction, and its normal has zero dot product with the given line direction.



Code reference -

`codes/C12/Q10.py`

2.2.5 Find where the line through $A(3,4,1)$ and $B(5,1,6)$ meets the plane through $P(2,1,2)$, $Q(3,1,0)$ and $R(4,-2,1)$.

Solution:

The two plane-edge vectors are

$$Q - P = (1, 0, -2).$$

$$R - P = (2, -3, -1).$$

Their cross product is proportional to $(2,1,1)$, so the plane equation is

$$2(x-2) + (y-1) + (z-2) = 0 \implies 2x + y + z = 7.$$

The line direction is $B - A = (2, -3, 5)$. Its affine representation is

$$\mathbf{r} = \mathbf{A} + t\mathbf{d}. \tag{2.1.5.1}$$

Thus

$$\mathbf{r} = (3,4,1) + t(2,-3,5).$$

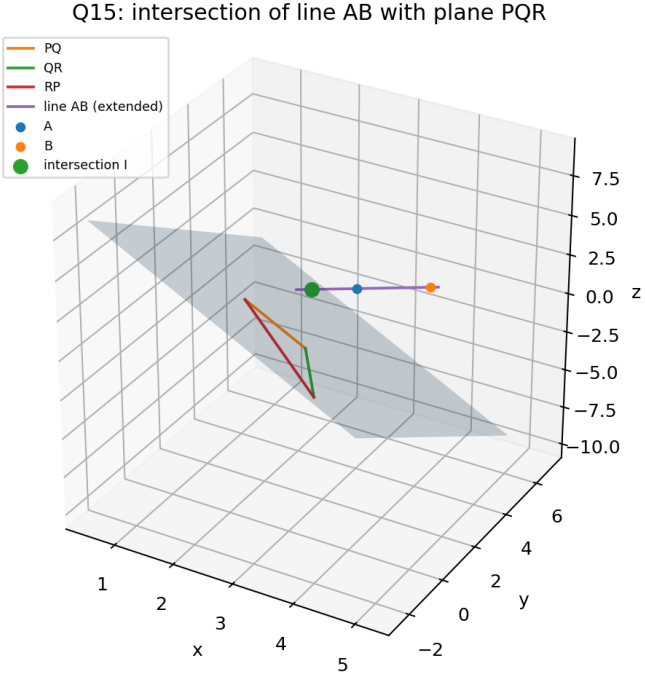
Substitution into the plane gives

$$2(3+2t)+(4-3t)+(1+5t)=7 \Longrightarrow 11+6t=7 \Longrightarrow t=-\frac{2}{3}.$$

The intersection point is therefore

$$\left(\frac{5}{3}, 6, -\frac{7}{3}\right).$$

Putting this point back into $2x+y+z$ gives $10/3+6-7/3=7$, which verifies the incidence condition.



Code reference -

`codes/C12/Q15.py`

2.2.6 If $y=\log\left(\frac{x}{a+bx}\right)^x$, prove that $x^3y''=(xy'-y)^2$.

Solution:

Interpret the displayed power as

$$y = x\log\left(\frac{x}{a+bx}\right),$$

on a domain where the logarithm is defined. The logarithmic derivative identity and

the Euler operator $\theta = x\,d/dx$ give

$$\frac{d}{dx}\log|u| = \frac{u'}{u}. \tag{2.1.6.1}$$

$$\theta y - y = xy' - y \tag{2.1.6.2}$$

Differentiate y explicitly:

$$y' = \log\left(\frac{x}{a+bx}\right) + x\left(\frac{1}{x} - \frac{b}{a+bx}\right) = \log\left(\frac{x}{a+bx}\right) + \frac{a}{a+bx}.$$

Differentiating once more,

$$y'' = \frac{1}{x} - \frac{b}{a+bx} - \frac{ab}{(a+bx)^2} = \frac{a^2}{x(a+bx)^2}.$$

The Euler combination simplifies without the logarithm:

$$xy' - y = x\frac{a}{a+bx} = \frac{ax}{a+bx}.$$

Squaring it and comparing with y'' gives

$$(xy'-y)^2 = \frac{a^2x^2}{(a+bx)^2} = x^3\frac{a^2}{x(a+bx)^2} = x^3y''.$$

Hence

$$x^3y'' = (xy'-y)^2.$$

2.2.7 If $A = \begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$, find $\det(A^n)$ for natural n.

Solution:

The symmetric matrix has eigenvectors $q_+ = (1,1)^T$ and $q_- = (1,-1)^T$. Their eigenvalues are

$$\lambda_+ = \cos\theta + \sin\theta.$$

$$\lambda_- = \cos\theta - \sin\theta.$$

Spectral decomposition therefore gives $\det A = \lambda_+\lambda_-$ and raises each eigenvalue to the nth power. Using the determinant-power identity,

$$\det(A^n) = (\det A)^n. \tag{2.1.7.1}$$

We obtain

$$\det(A^n) = (\lambda_+\lambda_-)^n = \big((\cos\theta+\sin\theta)(\cos\theta-\sin\theta)\big)^n = (\cos 2\theta)^n.$$

2.2.8 Find the inverse of $\begin{pmatrix}0 & 1 & 2\\ 1 & 2 & 3\\ 3 & 1 & 0\end{pmatrix}$ using elementary row operations.

Solution:

Let

$$A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 0 \end{pmatrix}.$$

Swap the first two rows to obtain a unit pivot, then eliminate below the pivots. The exact factorization is

$$PA = LU.$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -5 & 1 \end{pmatrix}.$$

$$U = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

The inverse is the matrix X satisfying $AX = I$; this defining relation is

$$AA^{-1} = I. \tag{2.1.8.1}$$

Solving the three triangular systems for the columns of X gives

$$A^{-1} = \begin{pmatrix} 3 & -2 & 1 \\ -9 & 6 & -2 \\ 5 & -3 & 1 \end{pmatrix}.$$

For the consistency check, multiplication of the first row gives $(0,1,2)A^{-1} = (1,0,0)$, and the other two rows similarly give $(0,1,0)$ and $(0,0,1)$. Therefore

$$A^{-1} = \begin{pmatrix} 3 & -2 & 1 \\ -9 & 6 & -2 \\ 5 & -3 & 1 \end{pmatrix}.$$

Code reference -

`codes/C12/Q30.py`

2.2.9 If $a+b+c \neq 0$ and $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, prove that $a = b = c$.

Solution:

Let A be the cyclic matrix in the question and let $\omega = e^{-2\pi i/3}$, so $1 + \omega + \omega^2 = 0$. The three-point DFT basis diagonalizes A, with eigenvalues

$$\lambda_0 = a + b + c. \tag{2.1.9.1}$$

$$\lambda_1 = a + b\omega + c\omega^2. \tag{2.1.9.2}$$

$$\lambda_2 = a + b\omega^2 + c\omega. \tag{2.1.9.3}$$

The determinant factorization is

$$0 = \det A = -\lambda_0\lambda_1\lambda_2 = -(a+b+c)(a^2+b^2+c^2-ab-bc-ca). \qquad (2.1.9.4)$$

Because $a+b+c \neq 0$, the second factor must vanish. The required sum-of-squares identity is

$$a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}\big((a-b)^2+(b-c)^2+(c-a)^2\big). \qquad (2.1.9.5)$$

For real a, b, c, each square is nonnegative. Their sum can be zero only if

$$a-b=0, \qquad b-c=0, \qquad c-a=0.$$

Hence

$$a=b=c.$$

2.2.10 Discuss continuity and differentiability of $f(x) = |x| + |x-1|$ on $(-1, 2)$.

Solution:

The absolute-value characteristic changes branch at $x = 0$ and $x = 1$. Resolving the signs gives

$$f(x) = |x| + |x-1| = \begin{cases} 1-2x, & -1 < x < 0, \\ 1, & 0 \le x < 1, \\ 2x-1, & 1 \le x < 2. \end{cases}$$

At the first breakpoint,

$$\lim_{x\to 0^-} f(x) = 1 = f(0) = \lim_{x\to 0^+} f(x),$$

and at the second,

$$\lim_{x\to 1^-} f(x) = 1 = f(1) = \lim_{x\to 1^+} f(x).$$

Thus the function is continuous throughout $(-1, 2)$. Its one-sided derivatives are

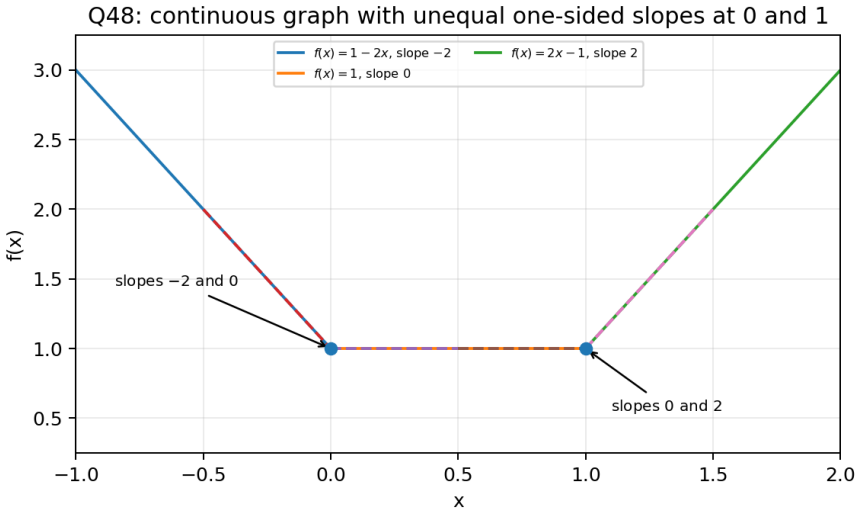
$$f'_-(0) = -2.$$

$$f'_+(0) = 0.$$

$$f'_-(1) = 0.$$

$$f'_+(1) = 2.$$

The slopes disagree at both breakpoints, so f is not differentiable at 0 or 1, but is differentiable elsewhere. Every branch has slope magnitude at most 2, hence the piecewise map is Lipschitz with constant 2.



Code reference -

`codes/C12/Q48.py`

2.2.11 On $\mathbb{R}$, define $(a,b) \in R$ when $a-b+\sqrt{3}$ is irrational. Determine whether R is reflexive, symmetric and transitive.

Solution:

Write the relation as the predicate $R(a,b)$: the number $a-b+\sqrt{3}$ must lie outside $\mathbb{Q}$. Its defining condition is

$$R(a,b) \iff a-b+\sqrt{3} \text{ is irrational.}$$

For reflexivity, $a-a+\sqrt{3}=\sqrt{3}$ is irrational, so $R(a,a)$ holds for every a. Symmetry fails: with $(a,b)=(\sqrt{3},0)$, the forward expression is $2\sqrt{3}$ (irrational), but the reversed expression is 0 (rational). Transitivity fails for

$$a = 0.$$

$$b = \frac{\sqrt{3}}{2}.$$

$$c = \sqrt{3}.$$

Both $a-b+\sqrt{3}=\sqrt{3}/2$ and $b-c+\sqrt{3}=\sqrt{3}/2$ are irrational, while $a-c+\sqrt{3}=0$. Therefore

$$R \text{ is reflexive, but neither symmetric nor transitive.}$$

2.2.12 On $\mathbb{R}\times\mathbb{R}$, define $(a,b)*(c,d)=(a+c,b+d)$. Prove commutativity and associativity,

and find the identity and the inverse of $(3,-5)$.

Solution:

The operation is coordinatewise addition, so it is closed on $\mathbb{R}^2$. Its defining law is

$$(a,b)*(c,d)=(a+c,b+d). \tag{2.1.10.1}$$

For associativity,

$$((a,b)*(c,d))*(e,f)=(a+c+e,b+d+f)=(a,b)*((c,d)*(e,f)). \tag{2.1.10.2}$$

For commutativity,

$$(a,b)*(c,d)=(a+c,b+d)=(c+a,d+b)=(c,d)*(a,b). \tag{2.1.10.3}$$

The identity is $(0,0)$, and the inverse of $(3,-5)$ solves

$$(3,-5)*(u,v)=(3+u,-5+v)=(0,0),$$

so it is $(-3,5)$. Thus the structure is an Abelian group and

$$e=(0,0), \qquad (3,-5)^{-1}=(-3,5).$$

2.2.13 On $\{0,1,2,3,4,5,6\}$ define $a*b=a+b$ if $a+b<7$, and $a*b=a+b-7$ otherwise. Give the operation table and show that 0 is the identity and $7-a$ is the inverse for nonzero a.

Solution:

The piecewise definition is addition modulo 7:

$$a*b=(a+b) \bmod 7. \tag{2.1.11.1}$$

The Cayley table is

$*$	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

The first row and column show $0*a=a*0=a$. For nonzero a, the residue $7-a$ lies in the set and

$$a*(7-a)=a+7-a\equiv 0 \pmod 7. \tag{2.1.11.2}$$

Since 7 is prime, every nonzero residue generates the cyclic group $\mathbb{Z}_7$. Therefore

$$0 \text{ is the identity and } a^{-1}=7-a \pmod 7.$$

2.2.14 Find any point of discontinuity of $f(x)=x-|x-x^2|$, $x\in[-1,1]$.

Solution:

Factor the expression inside the absolute value:

$$x - x^2 = x(1 - x). \tag{2.1.12.1}$$

On $[-1, 1]$, this quantity is nonnegative for $0 \le x \le 1$ and nonpositive for $-1 \le x \le 0$. The piecewise nonlinear map is therefore

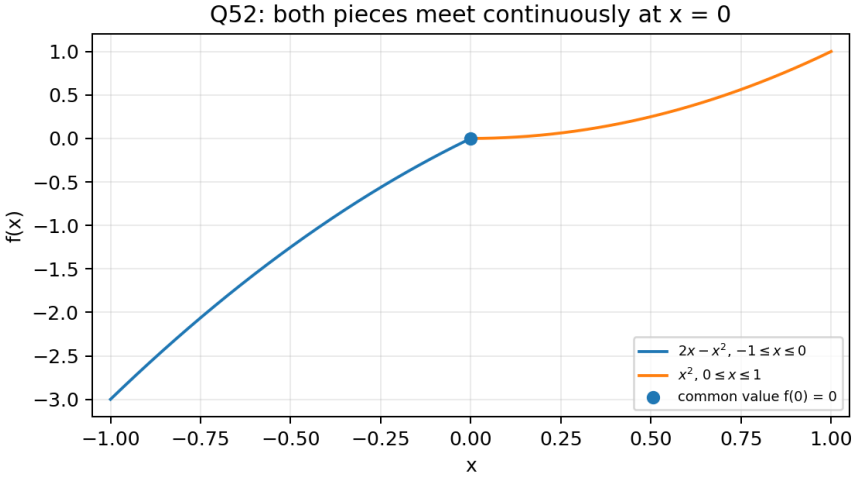
$$f(x) = \begin{cases} 2x - x^2, & -1 \le x \le 0, \\ x^2, & 0 \le x \le 1. \end{cases}$$

At the switching point, both branches give $f(0) = 0$. Moreover,

$$\lim_{x \to 0^-} (2x - x^2) = 0 = f(0) = \lim_{x \to 0^+} x^2.$$

Thus there is no discontinuity on the domain. The branch derivatives $2 - 2x$ and $2x$ are bounded in magnitude by 4, providing the stated Lipschitz consistency check. Hence

there is no point of discontinuity on $[-1, 1]$.



Code reference -

codes/C12/Q52.py

2.2.15 A man steps forward with probability 0.4 and backward with probability 0.6. Find the probability that after 5 steps he is one step away from his starting point.

Solution:

Encode a forward step by z and a backward step by z^{-1}. The one-step bilateral Laurent PGF and its five-step product are

$$G_5(z) = (0.4z + 0.6z^{-1})^5.$$

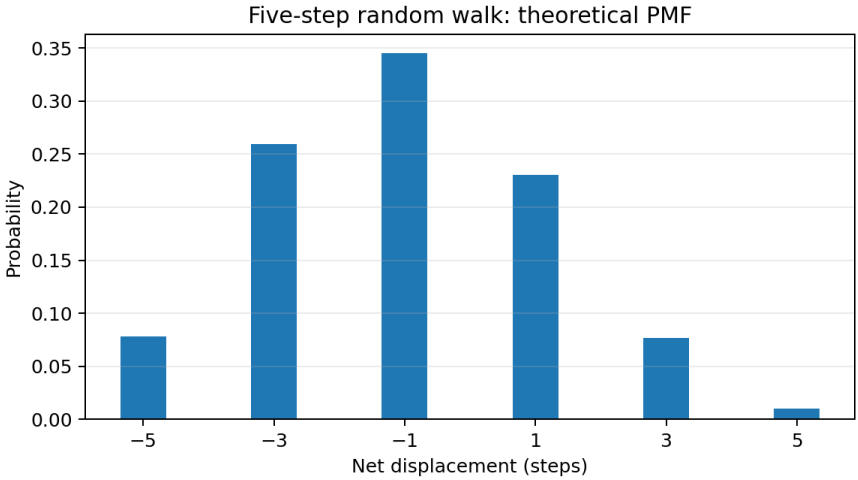
After five steps, displacement $+1$ requires three forward and two backward steps, while displacement -1 requires two forward and three backward steps. Extracting the two Laurent coefficients gives

$$P(|D| = 1) = \binom{5}{3}(0.4)^3(0.6)^2 + \binom{5}{2}(0.4)^2(0.6)^3.$$

Numerically,

$$\binom{5}{3}(0.064)(0.36) + \binom{5}{2}(0.16)(0.216) = 0.3456 + 0.2304 = 0.576.$$

The two cases are disjoint and exhaust the requested one-step displacement events.



Code reference -

codes/C12/Q53.py

2.2.16 A girl throws a die. If it shows 1 or 2, she tosses a coin three times; otherwise she tosses it once. Given that exactly one tail was obtained, find the probability that the die showed 3, 4, 5 or 6.

Solution:

Let H_1 denote die outcomes 1 or 2, H_2 denote outcomes $3, 4, 5, 6$, and let E be the event of exactly one tail. Then

$$P(H_1) = \frac{1}{3}.$$

$$P(H_2) = \frac{2}{3}.$$

Under H_1, exactly one tail in three tosses has likelihood $3(1/2)^3 = 3/8$; under H_2,

exactly one tail in one toss has likelihood $1/2$. Bayes' rule is

$$P(H_2 \mid E) = \frac{P(E \mid H_2)P(H_2)}{P(E)}. \tag{2.1.13.1}$$

The evidence probability is

$$P(E) = \frac{3}{8}\cdot\frac{1}{3} + \frac{1}{2}\cdot\frac{2}{3} = \frac{1}{8} + \frac{1}{3} = \frac{11}{24}.$$

Therefore

$$P(H_2 \mid E) = \frac{(1/2)(2/3)}{11/24} = \frac{1/3}{11/24} = \frac{8}{11}.$$

2.2.17 An urn contains 5 red and 2 black balls. Two balls are drawn without replacement. Let X be the number of black balls. State its possible values and find its mean and variance.

Solution:

Without replacement, X has a hypergeometric distribution with population size $N = 7$, black count $K = 2$, and sample size $n = 2$. Thus $X \in \{0, 1, 2\}$ and the probability mass is

$$P(X = 0) = \frac{10}{21}.$$

$$P(X = 1) = \frac{10}{21}.$$

$$P(X = 2) = \frac{1}{21}.$$

The three probabilities are

$$P(X = 0) = \frac{10}{21}, \qquad P(X = 1) = \frac{10}{21}, \qquad P(X = 2) = \frac{1}{21}.$$

The probability-generating function method uses

$$G_X(z) = \sum_x P(X = x) z^x. \tag{2.1.14.1}$$

Here

$$G_X(z) = \frac{10 + 10z + z^2}{21}.$$

$$G_X'(1) = \frac{12}{21} = \frac{4}{7}.$$

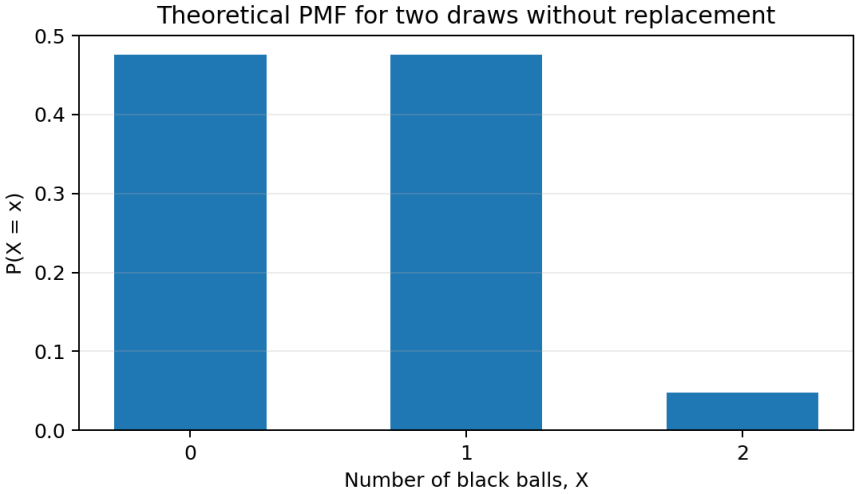
and

$$G_X''(1) = \frac{2}{21} = E[X(X-1)].$$

Consequently,

$$E[X^2] = E[X(X-1)] + E[X] = \frac{2}{21} + \frac{4}{7} = \frac{20}{21},$$

$$\operatorname{Var}(X) = E[X^2] - E[X]^2 = \frac{20}{21} - \frac{16}{49} = \frac{50}{147}.$$



Code reference -

`codes/C12/Q55.py`

2.2.18 In 3 trials of a binomial distribution, the probability of exactly 2 successes is 9 times the probability of 3 successes. Find the success probability.

Solution:

For three independent trials with success probability p, the binomial PGF is

$$G(z) = (1 - p + pz)^3. \tag{2.1.15.1}$$

The coefficients of z^2 and z^3 are

$$P(X = 2) = \binom{3}{2} p^2(1 - p) = 3p^2(1 - p).$$

$$P(X = 3) = p^3.$$

The given relation produces

$$3p^2(1 - p) = 9p^3 \implies 3p^2(1 - 4p) = 0.$$

Both roots are admissible probabilities in $[0, 1]$:

$$p = 0 \quad \text{or} \quad p = \frac{1}{4}.$$

The $p = 0$ case makes both probabilities zero, so it is retained as an edge case rather than divided away.

Code reference -

`codes/C12/Q56.py`

2.2.19 An urn contains 3 red and 5 black balls. A ball is drawn, its colour recorded and returned, and 2 more balls of that colour are added. Then two balls are drawn without replacement. Find the probability both are red.

Solution:

Condition on the colour of the reinforced first draw. The initial state is $(R, B) = (3, 5)$, so the branch probabilities are $3/8$ and $5/8$. After a red first draw the state is $(5, 5)$; after a black first draw it is $(3, 7)$. The law of total probability is

$$P(E) = \sum_i P(E \mid S_i)P(S_i). \tag{2.1.16.1}$$

For the red branch,

$$P(RR \mid R_1) = \frac{5}{10}\frac{4}{9} = \frac{2}{9}.$$

For the black branch,

$$P(RR \mid B_1) = \frac{3}{10}\frac{2}{9} = \frac{1}{15}.$$

Combining the mutually exclusive branches,

$$P(RR) = \frac{3}{8}\frac{2}{9} + \frac{5}{8}\frac{1}{15} = \frac{1}{12} + \frac{1}{24} = \frac{1}{8}.$$

The state update is applied before the two final draws, preserving the reinforcement dependence.

Code reference -

codes/C12/Q57.py

2.2.20 Solve graphically: minimise $z = 3x+5y$ subject to $x+2y \geq 10$, $x+y \geq 6$, $3x+y \geq 8$, $x, y \geq 0$.

Solution:

The feasible region is the intersection of three half-planes in the first quadrant. Its relevant corner is obtained by making the first two constraints active:

$$x + 2y = 10.$$

$$x + y = 6.$$

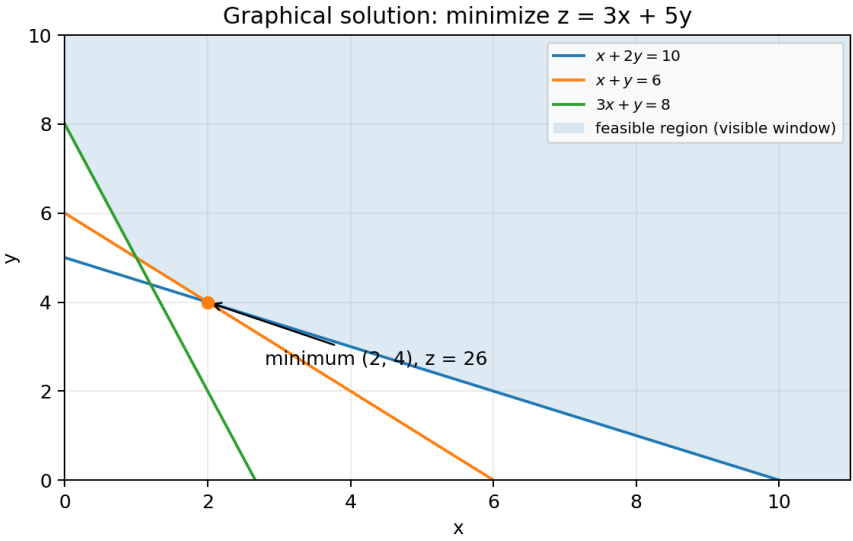
The objective admits the duality decomposition

$$3x + 5y = 2(x + 2y) + (x + y).$$

For every feasible point this gives $3x + 5y \geq 2(10) + 6 = 26$. At $(2, 4)$ equality holds, and the third constraint is satisfied because $3(2) + 4 = 10 \geq 8$. Therefore the graphical minimum is

$$(x, y) = (2, 4), \qquad z_{\min} = 3(2) + 5(4) = 26.$$

Equivalently, the dual certificate $(u, v, w) = (2, 1, 0)$ is feasible and has objective value $20 + 6 = 26$, confirming global optimality.



Code reference -

codes/C12/Q59.py

2.2.21 A dealer has Rs. 57,600 and space for at most 20 sewing machines. An electronic machine costs Rs. 3600 and earns Rs. 220 profit; a manual machine costs Rs. 2400 and earns Rs. 180 profit. How should he invest to maximise profit?

Solution:

Let e and m be the numbers of electronic and manual machines. Divide the budget constraint by Rs. 1200 to obtain

$$3e + 2m \leq 48.$$

$$e + m \leq 20.$$

$$e, m \geq 0.$$

The primal objective is $P = 220e + 180m$. The complementary-slackness equations for a tight optimum are

$$3u + v = 220.$$

$$2u + v = 180.$$

Subtracting gives $u = 40$, and then $v = 100$. The dual value is

$$48u + 20v = 48(40) + 20(100) = 3920.$$

The primal intersection of the two resource boundaries satisfies

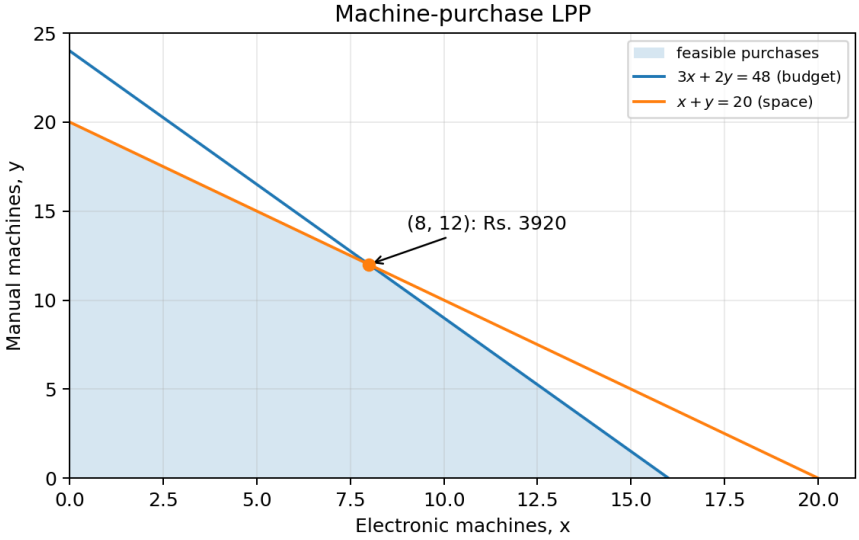
$$3e + 2m = 48, \qquad e + m = 20 \Longrightarrow e = 8, \quad m = 12.$$

Its profit is

$$3e + 2m = 48.$$

$$e + m = 20.$$

Both resource constraints are tight, and primal and dual values agree, so complementary slackness certifies the allocation $(e, m) = (8, 12)$.



Code reference -

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codes/C12/Q60.py
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2.2.22 A tangent to $x^2 + y^2 = 4$ at a point in the first quadrant meets the positive axes at A and B. Find the minimum of $OA + OB$.

Solution:

Parameterize the point of tangency by $(2\cos\theta, 2\sin\theta)$ with $0 < \theta < \pi/2$. The tangent and its intercepts follow from the constrained geometry:

$$\tan\theta: \quad x\cos\theta + y\sin\theta = 2.$$

Setting $y = 0$ and $x = 0$ gives intercept lengths $OA = 2\sec\theta$ and $OB = 2\csc\theta$. Hence

$$F(\theta) = OA + OB = 2\sec\theta + 2\csc\theta.$$

Stationarity gives

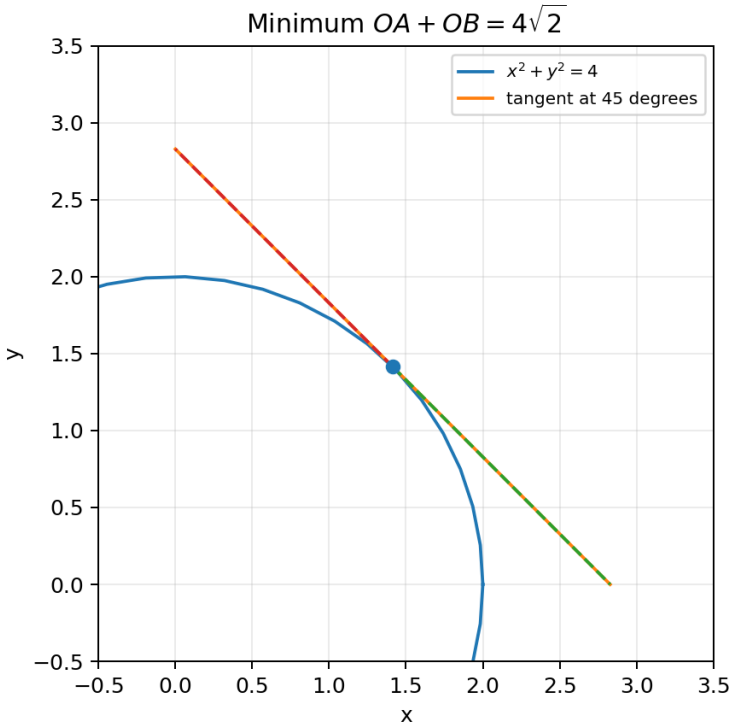
$$F'(\theta) = 2\sec\theta\tan\theta - 2\csc\theta\cot\theta = 0.$$

After multiplying by positive factors in the first quadrant,

$$\sin^3\theta = \cos^3\theta \Longrightarrow \theta = \frac{\pi}{4}.$$

The objective tends to infinity at either open endpoint, so this stationary point is the global minimum. Therefore

$$F_{\min} = 2\sqrt{2} + 2\sqrt{2} = 4\sqrt{2}.$$



Code reference -

codes/C12/Q64.py

2.2.23 Find the maximum area of an isosceles triangle inscribed in $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with its vertex at an end of the major axis.

Solution:

Fix the vertex at $(4, 0)$ and use symmetry for the base points:

$$P = (u, v).$$

$$Q = (u, -v).$$

The ellipse constraint gives $v = 3\sqrt{1 - u^2/16}$, with $-4 \le u \le 4$. The base has length $2v$ and the horizontal altitude is $4 - u$, so the constrained area objective is

$$A(u) = 3(4 - u)\sqrt{1 - u^2/16}, \qquad -4 \le u \le 4.$$

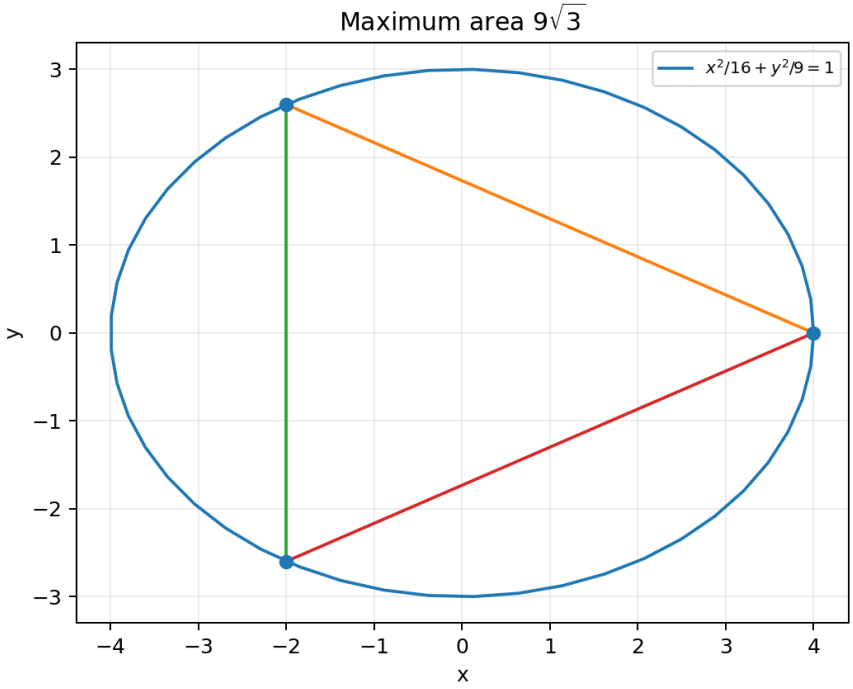
To locate interior extrema, maximize the square of the nonnegative area. Differentiating the equivalent expression

$$A(u)^2 = 9(4 - u)^2 \left(1 - \frac{u^2}{16}\right)$$

gives the stationary condition $u = -2$ (the other factor corresponds to a zero-area endpoint). The endpoints $u = \pm 4$ give $A = 0$, while at the interior point

$$A(-2) = 3(4 + 2)\sqrt{1 - \frac{4}{16}} = 18\sqrt{\frac{3}{4}} = 9\sqrt{3}.$$

Since the feasible interval is compact and the endpoints have zero area, this positive stationary value is the global maximum.



Code reference -

codes/C12/Q65.py